



**P. A. COLLEGE OF ENGINEERING AND
TECHNOLOGY**
(An Autonomous Institution)



Department of Artificial Intelligence And Data Science

ALZHEIMER'S DISEASE DETECTION USING MRI IMAGE CLASSIFICATION WITH AN ENSEMBLE DEEP LEARNING MODEL

PRESENTED BY,

Adhithyan R(721722243001)

Ajaykumar M(721722243003)

Raajukamal A(721722243032)

Sivanandham SB(721722243040)

GUIDED BY,

Mr. M. Sabarivel, M.Tech., AP/IT

AGENDA

- OBJECTIVE
- INTRODUCTION
- LITERATURE SURVEY
- ALZHEIMER'S DISEASE PREDICTION WITH MULTIMODAL DATA
- PROPOSED SOLUTION
- SYSTEM ARCHITECTURE
- MODULES
- DATA COLLECTION
- RESULT & OUTPUT
- REFERENCES

OBJECTIVE

- To design and develop an **MRI-based Alzheimer's disease detection system** using an **ensemble deep learning architecture**.
- To combine the strengths of **Custom CNN** and **pretrained ResNet50** for robust feature extraction.
- To improve **classification accuracy and generalization** by fusing shallow and deep features
- To assist doctors with **computer-aided early diagnosis** and reduce manual diagnostic errors.

INTRODUCTION

- MRI brain scans capture subtle structural changes in the brain that are strongly associated with Alzheimer's disease. These changes may not be easily detectable through routine clinical examination, but AI-based image analysis can identify them earlier.
- Most existing Alzheimer's detection systems depend on clinical test results, medical history, or psychological assessments. These methods may delay diagnosis, require multiple hospital visits, and can become unreliable when patient data is missing or incomplete.
- In our approach, an ensemble deep learning model combining a Custom CNN and ResNet50 is used to automatically learn and fuse features from MRI brain images to classify Alzheimer's disease. This system acts as a computer-aided diagnostic tool, supporting doctors in early detection, improved accuracy, and effective treatment planning.

LITERATURE SURVEY

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2025	Sobhana Jahan, Md. Rawnak Saif Adib, Syed Mahmudul Huda, Md. Sazzadur Rahman, M. Shamim Kaiser, A. S. M. Sanwar Hosen, Deepak Ghimire, Mi Jin Park	Federated Explainable AI-based Alzheimer's Disease Prediction With Multimodal Data	Random Forest (RF), Federated Learning (FL), SHAP (XAI), Multimodal data (Clinical, MRI Segmentation, Psychological)	Requires multiple data modalities (clinical, psychological, MRI), complex data preprocessing, manual feature selection using Pearson correlation, time-consuming data collection, high computational overhead

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2024	J Faritha Banu, Kingsly Stephen R, N Aditya, Dhanush Raaghav L C	Alzheimer's Disease Detection using Deep Learning Algorithm	Convolutional Neural Network (CNN), OASIS dataset	Low precision values (0.209- 0.255), requires large balanced datasets using SMOTE, manual interpretation still needed, limited to MRI modality only

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2022	Duaa AlSaeed, Samar Fouad Omar	Brain MRI Analysis for Alzheimer's Disease Diagnosis Using CNN- Based Feature Extraction and Machine Learning	CNN, SVM, Random Forest, Softmax classifier	Requires extensive preprocessing (skull removal, smoothing, resizing), limited to single imaging modality (MRI), high computational resources needed, dataset size limitations affect performance

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2020	V. P. S. Rallabandi, K. Tulpule, M. Gattu	Automatic classification of cognitively normal, mild cognitive impairment and Alzheimer's disease using structural MRI	FreeSurfer analysis, Non-linear SVM (RBF kernel), Naive Bayesian, K-NN, Random Forest, Decision Tree	Moderate accuracy (75%), requires FreeSurfer preprocessing, limited to cortical thickness features (68 features), computationally intensive feature extraction

Contd.,

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2020	Baiying Lei, Yun Zhu, Enmin Liang, Peng Yang, Shaobin Chen, Huoyou Hu, Haoran Xie, Ziyi Wei, Fei Hao, Xuegang Song, et al.	Federated domain adaptation via transformer for multi-site alzheimer's disease diagnosis	CNN, SVM	Requires multiple evaluation processes for different segmentation methods, dataset size limitations, computationally intensive

ALZHEIMER'S DISEASE PREDICTION WITH MULTIMODAL DATA

- Most existing systems for Alzheimer's prediction **make use of clinical and psychological data such as patient history, memory test scores, genetic details, laboratory reports, and other health records.**
- These systems commonly apply **Machine Learning techniques like Random Forest, SVM, and Logistic Regression** to classify the disease stages.
- Since these methods mainly rely on multimodal tabular data, they require multiple hospital visits, medical tests, and continuous record maintenance.
- If any information is missing or incomplete, the system performance reduces significantly. Also, these systems generally do not process MRI images directly, so deeper structural brain changes are not fully captured.

LIMITATIONS OF EXISTING SYSTEMS

- **System metrics not measured** - Training time, computation cost, communication overhead not calculated
- **Limited to OASIS-3 dataset** - Needs validation on other diverse datasets
- **Complex preprocessing** - Manual feature selection and data imputation required
- **High computational overhead** - FL implementation challenges not fully addressed
- **No client-specific explainability** - XAI focused only on server-side global model
- **Limited generalizability** - Tested only on OASIS-3 with five classes
- **Subjective thresholds** - Pearson correlation thresholds (0.75-0.95) manually determined

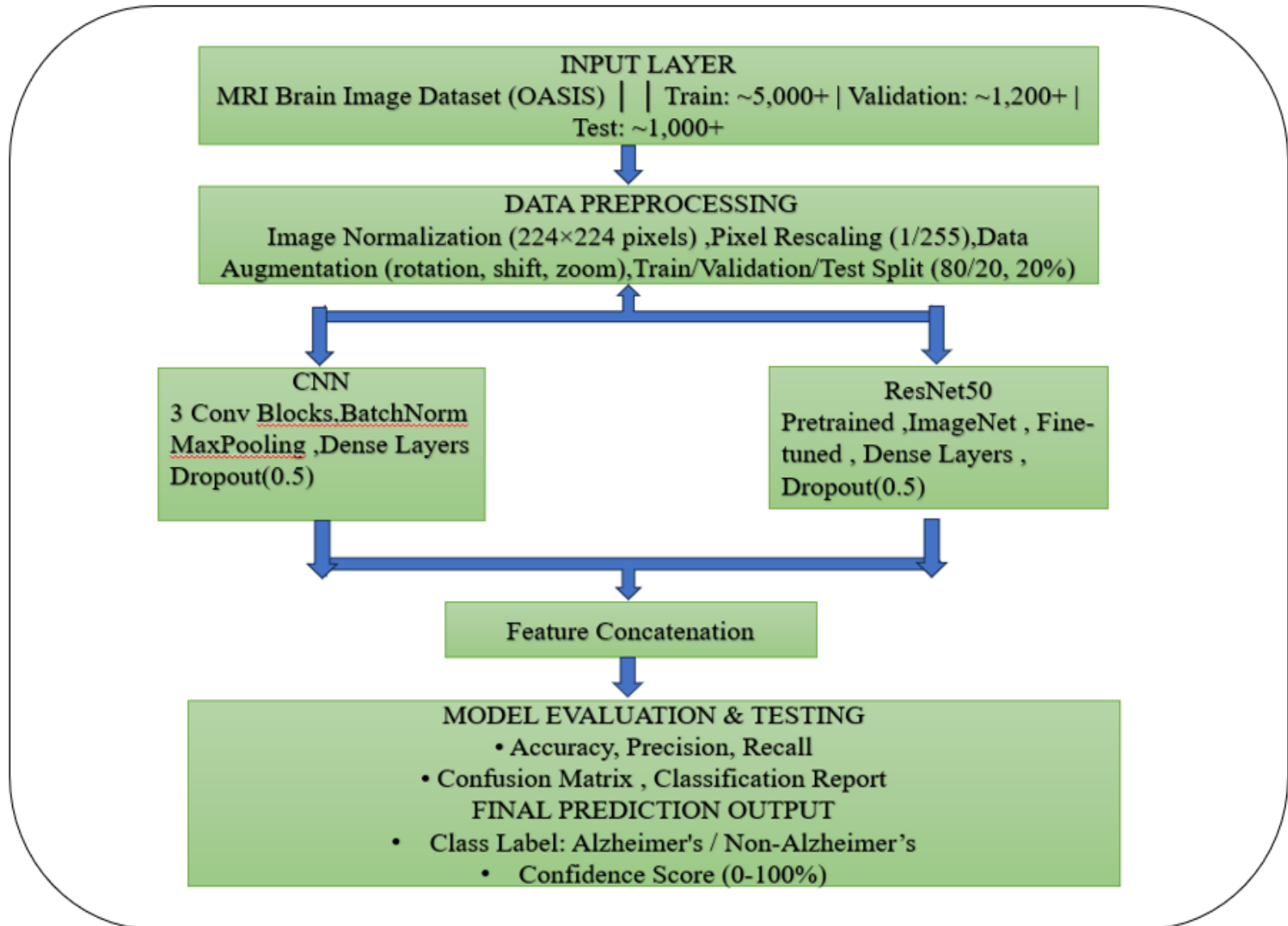
DRAWBACK

- **Not suitable for early-stage Alzheimer's detection**
 - Existing systems fail to identify subtle brain changes in the early stages, leading to delayed diagnosis.
- **Requires multiple hospital visits and tests**
 - Diagnosis depends on repeated clinical and psychological evaluations, increasing time, cost, and patient burden.
- **Performance drops with incomplete patient data**
 - System accuracy reduces significantly when patient records or test results are missing or inconsistent.
- **High dependency on clinical expertise**
 - Results rely heavily on manual interpretation by doctors, increasing subjectivity and human error.
- **No automated MRI-based decision support**
 - MRI brain images are not directly analyzed using deep learning, limiting accurate and early detection.

ALZHEIMER'S DISEASE PREDICTION USING MRI IMAGE CLASSIFICATION WITH AN ENSEMBLE DEEP LEARNING MODEL

- The proposed system introduces an **Enhanced Hybrid Ensemble Model** that combines:
 - A **Custom Convolutional Neural Network (CNN)** to extract low-level spatial and texture features from MRI images.
 - A **ResNet50 deep network**, pretrained on ImageNet and fine-tuned on MRI data, to capture **high-level structural brain abnormalities**.
- Both feature sets are **concatenated at feature level** and passed to a shared classification head.
- This ensemble approach:
 - Improves robustness
 - Reduces overfitting
 - Achieves better performance than individual models

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE

➤ Input Layer

- The system takes **MRI brain scan images** as input from a standard Alzheimer's MRI dataset (such as OASIS).
- The dataset is divided into **training, validation, and testing sets** to ensure proper model learning and unbiased performance evaluation.

➤ Data Preprocessing

Before feeding the images into the deep learning models, preprocessing is performed to enhance image quality and consistency:

- MRI images are resized to **224×224 pixels**
- Pixel values are normalized and rescaled (1/255)
- **Data augmentation** techniques such as rotation, shifting, and zooming are applied
- The dataset is split into **training, validation, and test sets**

➤ **Feature Extraction Using CNN**

- A **Custom Convolutional Neural Network (CNN)** is used to extract features:
 - Multiple convolution blocks learn low-level and mid-level brain features
 - **Batch Normalization** improves training stability
 - **Max Pooling** reduces spatial dimensions
 - **Dropout (0.5)** prevents overfitting
 - Dense layers convert extracted features into meaningful representations
- This branch learns **structural and texture patterns** related to Alzheimer's disease.

➤ **Feature Extraction Using ResNet50**

- In parallel, a **ResNet50 model** is used:
 - Pretrained on **ImageNet**
 - Fine-tuned using MRI brain images
 - Uses **residual connections** for deep feature learning
 - Dense layers and dropout improve classification accuracy.

➤ **Feature Fusion (Ensemble Layer)**

- Features extracted from **CNN and ResNet50** are **concatenated**
- This **feature-level ensemble** combines shallow and deep brain representations for improved performance.

➤ **Model Evaluation & Testing**

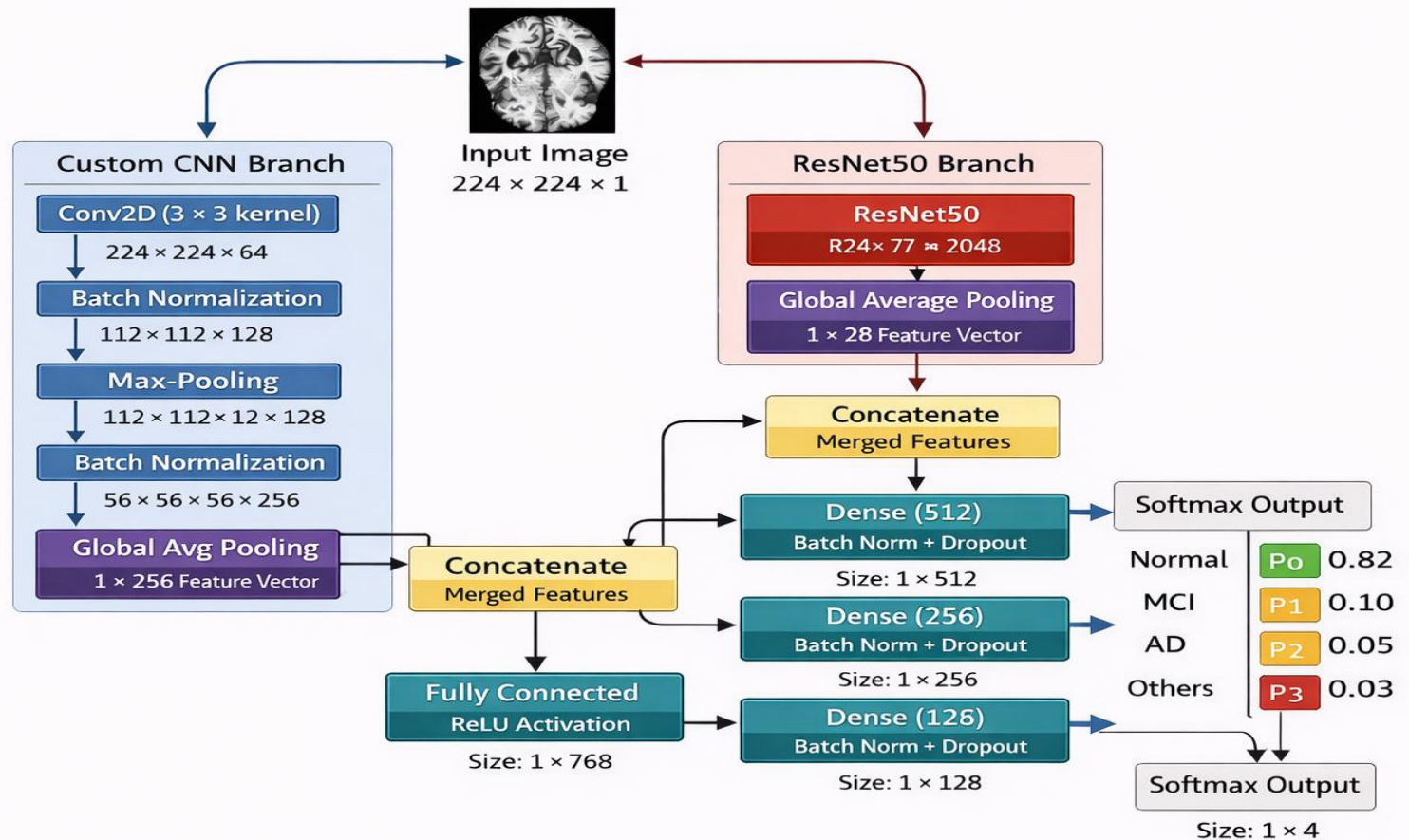
- The ensemble model is evaluated using standard performance metrics:
- Accuracy , Precision , Recall , Confusion Matrix , Classification Report
- This ensures **reliable and unbiased assessment** of model performance.

➤ **Final Prediction Output**

- The system produces the final diagnosis result:
 - **Class Label:** Alzheimer's / Non-Alzheimer's (or multi-class stages)
 - **Confidence Score:** 0–100%
- The output acts as a **computer-aided diagnostic support tool** for clinicians.

MODEL ARCHITECTURE – ENHANCED HYBRID

Enhanced Hybrid Model



MODEL ARCHITECTURE – ENHANCED HYBRID

➤ Input Layer

The system takes a **brain MRI image** as input with a size of $224 \times 224 \times 1$, where the image is in **grayscale format**. This MRI image contains essential **structural and anatomical brain information**, which is crucial for identifying patterns associated with **Alzheimer's disease and cognitive impairment**.

➤ CNN Branch

First Convolution Layer (Conv_1)

- A **convolution operation** is applied using 3×3 kernels to extract **low-level features** such as:
 - Edges
 - Texture
 - Brain tissue intensity variations
- This layer generates multiple **feature maps**, enabling the network to learn basic visual patterns present in MRI scans.

➤ **Batch Normalization**

- Batch Normalization is applied after convolution to:
 - Stabilize learning
 - Speed up convergence
 - Reduce internal covariate shift
- This improves overall training efficiency and model generalization.

➤ **Max-Pooling Layer**

- A **2×2 max-pooling operation** is performed to:
 - Reduce spatial dimensions
 - Lower computational complexity
 - Retain the most dominant features
- Pooling helps the model become **translation-invariant** and more robust to small spatial variations.

➤ **Second Convolution Layer (Conv_2)**

- Another convolution layer with 3×3 **kernels** is applied to extract **higher-level features**, such as:
 - Structural abnormalities
 - Shape variations in brain regions
 - Cortical and ventricular changes related to Alzheimer's disease
- These features are more meaningful for medical diagnosis.

➤ **Repeated Convolution + Pooling Operations**

- The convolution and pooling steps are **repeated multiple times** to:
 - Learn deeper and more complex representations
 - Capture spatial relationships between different brain regions
 - Improve disease pattern recognition accuracy
- This hierarchical feature learning enhances the discriminative power of the CNN branch.

➤ **Global Average Pooling**

- Global Average Pooling is applied to:
 - Reduce each feature map to a single value
 - Minimize overfitting
 - Produce a compact and meaningful **feature vector**
- This results in a fixed-length feature representation from the Custom CNN branch.

➤ **ResNet50 Branch (Transfer Learning)**

In parallel, the same input image is fed into a **pretrained ResNet50 model**, which has been trained on the **ImageNet dataset**. ResNet50 uses **residual connections** that help:

- Prevent vanishing gradients
- Enable very deep feature learning
- This branch extracts **high-level semantic features** relevant to complex brain structures.

➤ **Global Average Pooling (ResNet Branch)**

- The output feature maps from ResNet50 are passed through **Global Average Pooling**, producing a **2048-dimensional feature vector** that summarizes deep spatial information.

➤ **Feature Fusion (Concatenation Layer)**

- The feature vectors obtained from:
 - Custom CNN branch
 - ResNet50 branch are **concatenated** into a single combined feature vector.

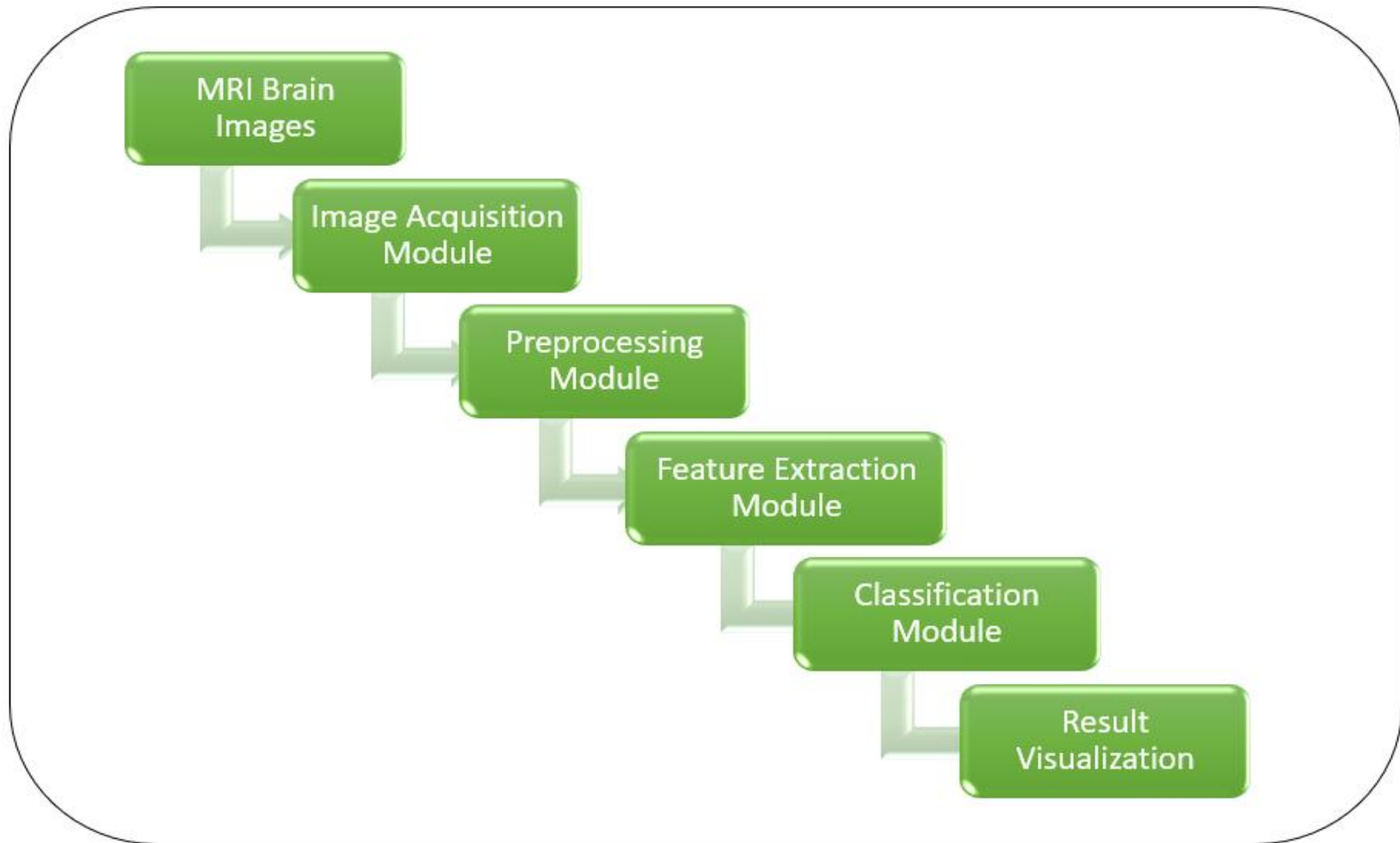
➤ **Fully Connected Layers (Classification Head)**

- **Dense Layer 1 (512 Neurons)**
 - The fused feature vector is passed through a **fully connected dense layer** with:
 - **ReLU activation** for non-linearity
 - **Batch Normalization** for stability
 - **Dropout** to prevent overfitting
 - This layer performs high-level reasoning on the combined features.
- **Dense Layer 2 (256 Neurons)**
 - A second dense layer further refines the learned features and improves class separability while maintaining regularization using Batch Normalization and Dropout.
- **Dense Layer 3 (128 Neurons)**
 - This layer compresses the feature representation and prepares it for final classification.

➤ **Softmax Output Layer**

- The final layer is a **Softmax classifier** with **4 output neurons**, corresponding to different classes such as:
 - Normal
 - Mild Cognitive Impairment (MCI)
 - Alzheimer's Disease (AD)
 - Other cognitive conditions
- The Softmax function outputs **probability values** for each class, and the class with the **highest probability** is selected as the final prediction.

MODULES



MODULES

- **Image Acquisition Module**
 - Loads MRI brain images from multiple Alzheimer's datasets.
- **Preprocessing Module**
 - Performs resizing, normalization, and augmentation.
- **Feature Extraction Module**
 - Uses **CNN and ResNet50 ensemble** to extract deep MRI features.
- **Feature Fusion Module**
 - Combines CNN and ResNet50 features using concatenation.
- **Classification Module**
 - Uses Softmax layer to classify Alzheimer's stages.
- **Result Visualization Module**
 - Displays prediction results, confidence score, and confusion matrix.

COMPLETE ADVANTAGE OVERVIEW

➤ **Simplicity & Accessibility**

- Uses a **single MRI modality** (no clinical or psychological data required)
- Eliminates dependence on **longitudinal patient records**
- Performs **direct MRI image analysis** using deep learning
- Easy to deploy as a **standalone computer-aided diagnostic system**

➤ **Data Processing**

- **Minimal preprocessing** (image resizing and normalization)
- **Automatic feature learning** using CNN and ResNet50
- No manual feature extraction or handcrafted feature selection
- **No feature engineering expertise required**

COMPLETE ADVANTAGE OVERVIEW

➤ **Computational Efficiency**

- Efficient training using **pretrained ResNet50**
- Faster convergence compared to training from scratch
- **Low inference latency** (suitable for near real-time diagnosis)
- GPU recommended for training, **CPU sufficient for deployment**

➤ **Scalability & Deployment**

- Can be easily deployed in **hospitals or diagnostic centers**
- Supports **web-based integration** using Flask or FastAPI
- Works as a **single-institution system** without data sharing
- Can scale horizontally using multiple inference servers

TECHNOLOGY STACK

- **Programming:** Python 3.8+
- **Algorithms:** CNN, ResNet50, BatchNormalization, Dropout, Softmax
- **Libraries:** TensorFlow/Keras, OpenCV, NumPy, Matplotlib, Seaborn, Pandas
- **Datasets:** Best Alzheimer's MRI Dataset - 4 classes, 7,200+ images
- **Hardware:** GPU (NVIDIA CUDA 11.x), RAM (8GB+), Laptop

DATA COLLECTION

- **Best Alzheimer's MRI Dataset**
 - Contains **7,200+ MRI brain scans** from **1,098 subjects**.
 - Includes **4 disease classes**: CN (Cognitively Normal), MCI (Mild Cognitive Impairment), AD (Alzheimer's Disease), Dementia.
 - Provides **longitudinal data** collected over **15+ years** with real-world clinical variations.
- **Sachin Kumar's Alzheimer's MRI Dataset**
 - Includes **3,000+ preprocessed MRI images** with balanced class distribution.
 - Captures **multiple MRI sequences** (T1, T2, FLAIR) with anatomical variations.
 - Useful for training models on **diverse imaging protocols** and scanner types.

DATA COLLECTION

- **Aryan Singhal's Augmented Alzheimer's Dataset**
 - Contains **10,000+ augmented MRI images** using data augmentation techniques.
 - Features **equal class representation** preventing class imbalance issues.
 - Includes **synthetic variations** (rotations, shifts, zooms) of original scans.
 - Enables model training with **enhanced generalization capability** across imaging variations.

RESULT & OUTPUT

=====

 EVALUATING MODEL ON TEST SET

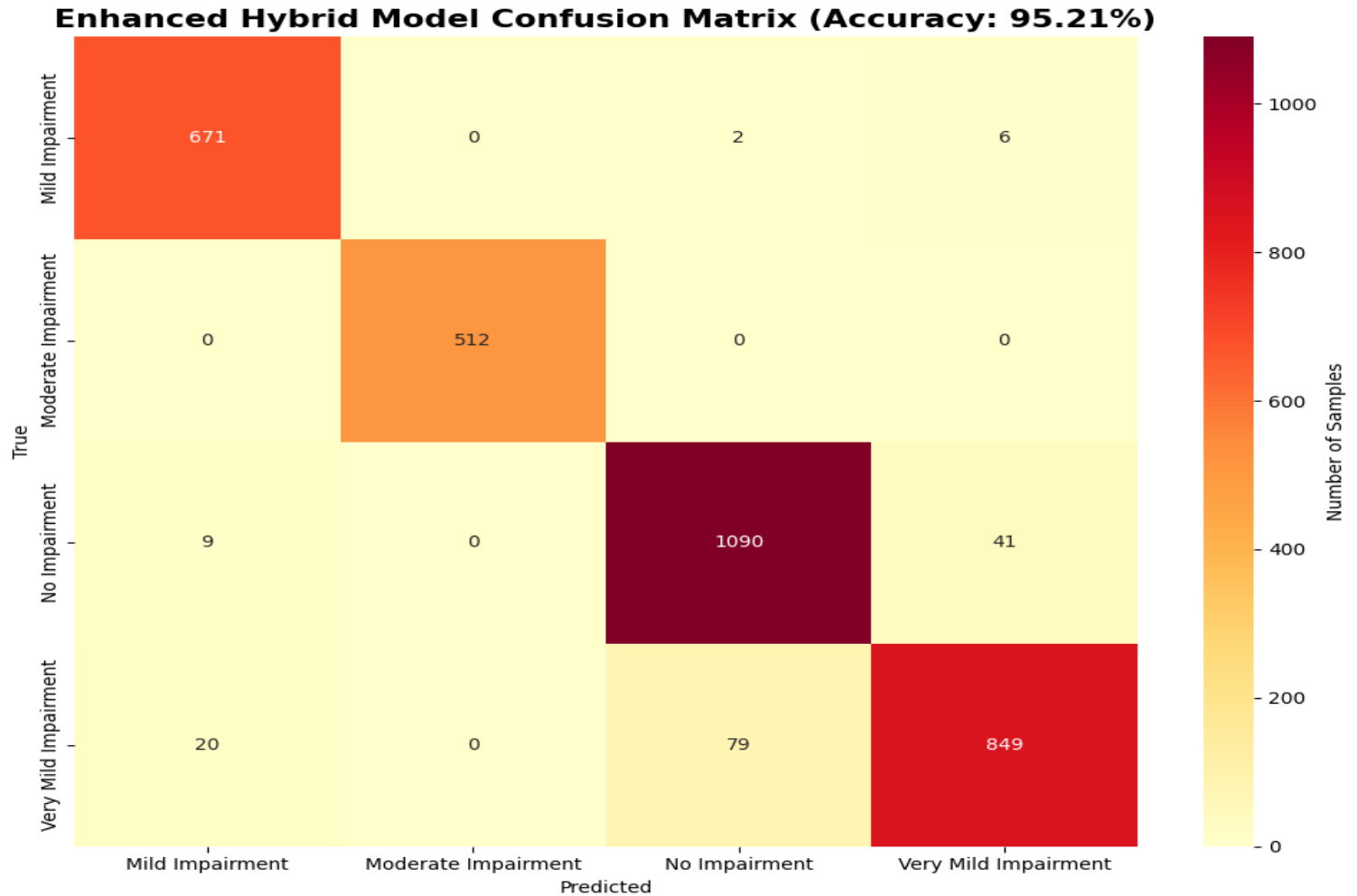
=====

103/103  18s 175ms/step - accuracy: 0.9794 - auc: 0.9984 - loss: 0.0689 - precision: 0.9799 - recall: 0.9778

 Test Performance:

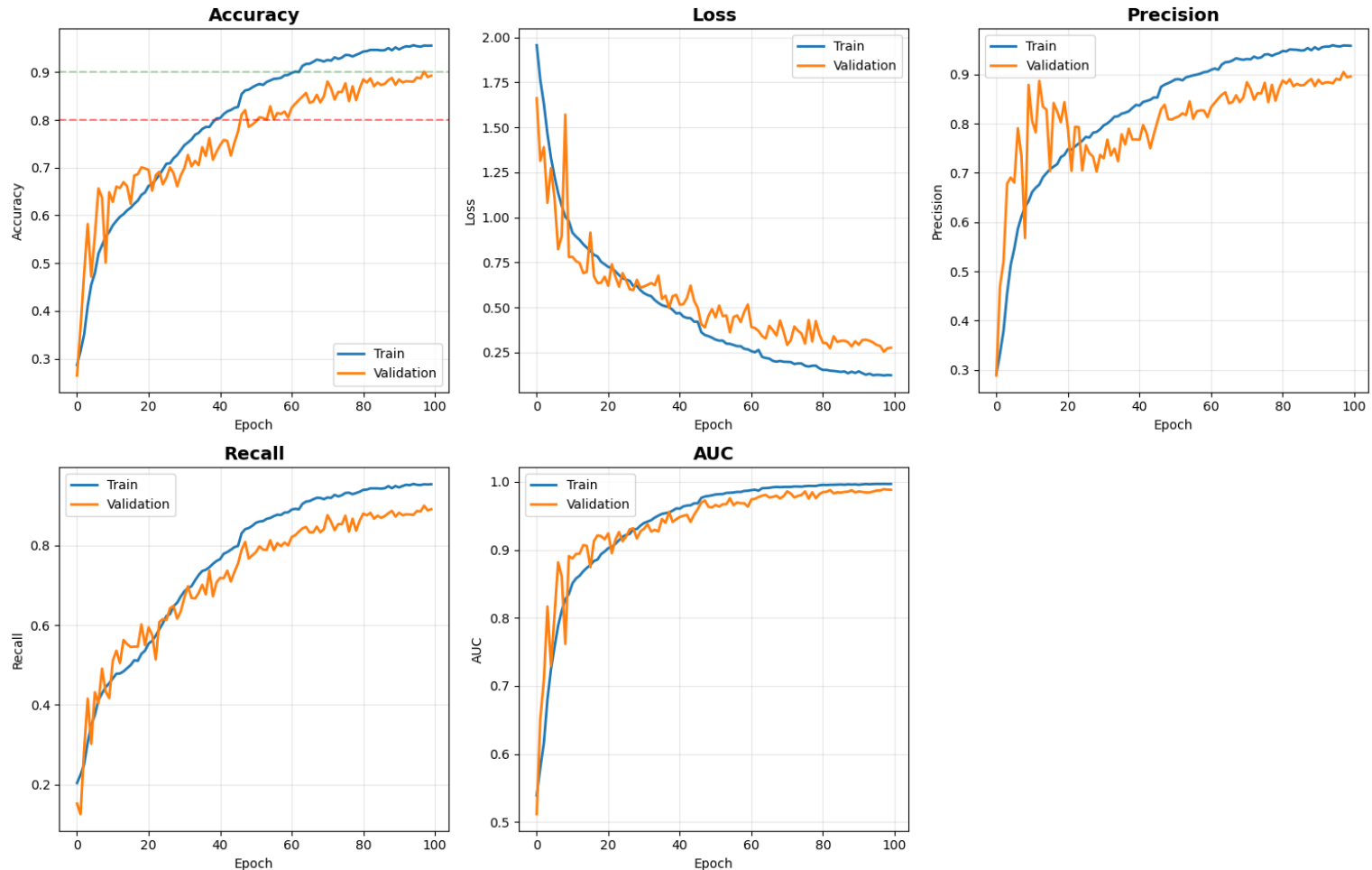
- Loss: 0.1415
- Accuracy: 0.9521 (95.21%)
- Precision: 0.9538
- Recall: 0.9503
- AUC: 0.9955

RESULT & OUTPUT



RESULT & OUTPUT

Training History with Combined Dataset



RESULT & OUTPUT

 **Alzheimer's MRI Analysis**
Enhanced Hybrid CNN + ResNet50 Model | 95.2% Accuracy

✓ 95.2% Accuracy

 21,519 MRI Scans

 Hybrid Model

 **Upload MRI Scan**

 Image


Drop Image Here
- or -
Click to Upload

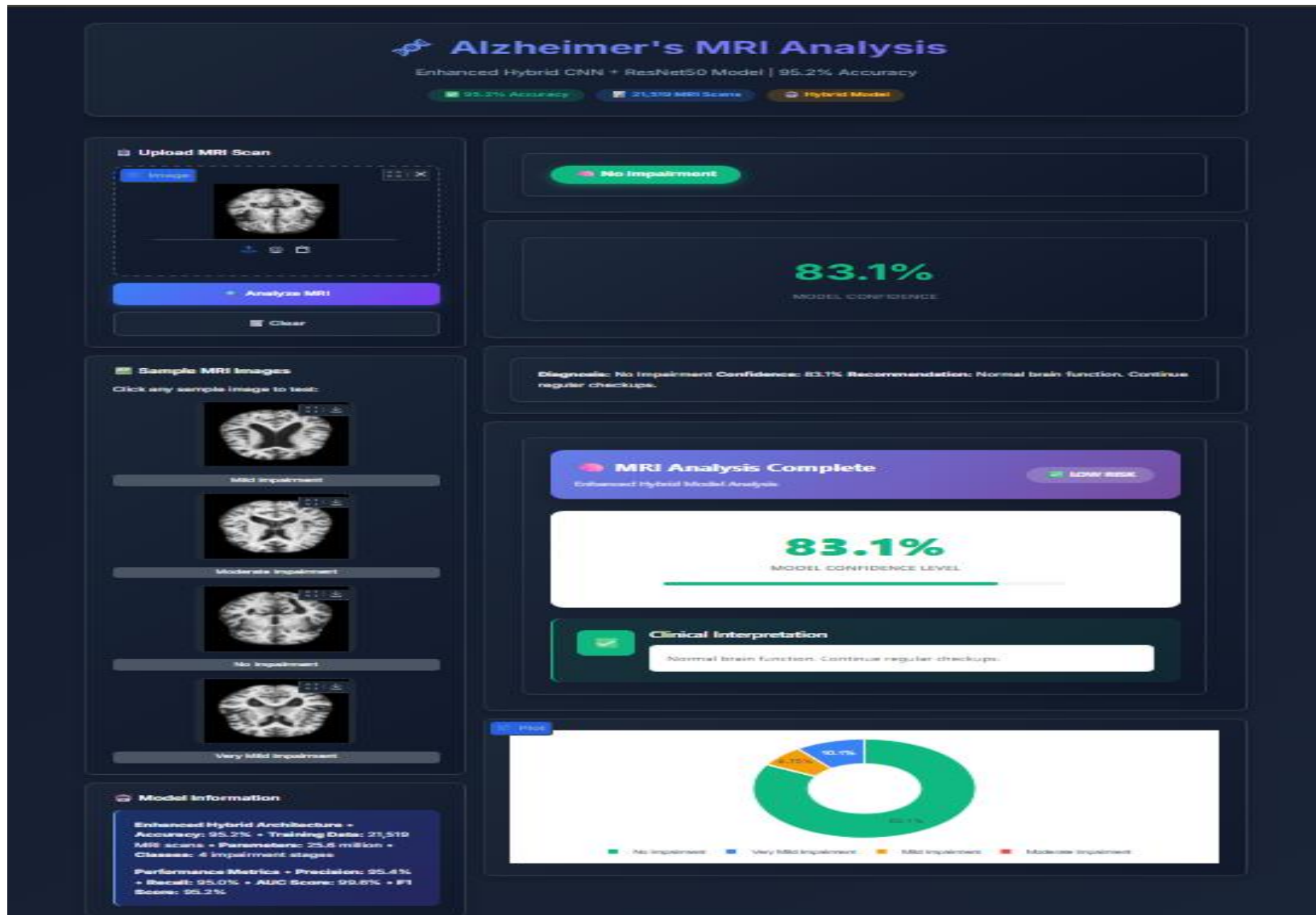
Analyze MRI

 Clear

Upload MRI for analysis...

Confidence Level
0.0%

RESULT & OUTPUT



REFERENCES

1. S. Jahan, Md. Rawnak Saif Adib, S. M. Huda, Md. S. Rahman, M. S. Kaiser, A. S. M. S. Hosen, D. Ghimire, and M. J. Park (2025),“*Federated Explainable AI-based Alzheimer’s Disease Prediction with Multimodal Data,*” IEEE Access.
2. J. Faritha Banu, Kingsly Stephen R., N. Aditya, and Dhanush Raaghav L. C. (2024),“*Alzheimer’s Disease Detection Using Deep Learning Algorithm,*” IJRESM Journal of Engineering and Technology.
3. Singh, S.K., Hussain, S., Kumar, A., Sinha, N., and Goyal, S. (2024) “*An Automated Deep Learning Model for Diagnosis of Alzheimer's Disease Using Deep Feature Fusion,*” IEEE International Conference on Computing Communication and Networking Technologies (ICCCNT), pp. 1-7.

4. Shamrat, F.J.M., Raihan, M., Hasan, T., Rahman, A., and Islam, M.R.(2023) ,"***AlzheimerNet: An Effective Deep Learning Based Proposition for Alzheimer's Disease Stages Classification from Functional Brain Changes in Magnetic Resonance Images***," IEEE Access, Vol. 11, pp. 16376-16395.
5. Chabib, C.M., Hadjileontiadis, L.J., and Al Shehhi, A.(2023), "***DeepCurvMRI: Deep Convolutional Curvelet Transform-Based MRI Approach for Early Detection of Alzheimer's Disease***," IEEE Access, Vol. 11, pp. 44650-44659.
6. Duan, Y., Wang, R., and Li, Y.(2023) ,"***Aux-VIT: Classification of Alzheimer's Disease from MRI Based on Vision Transformer with Auxiliary Branch***," 5th International Conference on Communications, Information System and Computer Engineering (CISCE), pp. 382-386.

7. V. P. S. Rallabandi, K. Tulpule, and M. Gattu (2020), ***“Automatic Classification of Cognitively Normal, Mild Cognitive Impairment and Alzheimer’s Disease Using Structural MRI,”*** IEEE Access.
8. Baiying Lei, Yun Zhu, Enmin Liang, Peng Yang, Shaobin Chen, Huoyou Hu, Haoran Xie, Ziyi Wei, Fei Hao (2020), ***“Federated Domain Adaptation via Transformer for Multi-site Alzheimer’s Disease Diagnosis,”*** IEEE Transactions on Medical Imaging.
9. Lei, B., Yun Zhu, Enmin Liang, Peng Yang, Shaobin Chen, Huoyou Hu, Haoran Xie, Ziyi Wei, and Fei Hao.(2020), ***"Federated Domain Adaptation via Transformer for Multi-site Alzheimer's Disease Diagnosis,"*** IEEE Transactions on Medical Imaging, Vol. 39, No. 12, pp. 4154-4163.

☺ **THANK YOU!!**